

$$\Rightarrow x \in J^n(A) \Leftrightarrow 1-xy \text{ is a unit in } A \quad \forall y \in A.$$

Prop: $J^n(A) := \{x \in A : 1-xy \text{ is a unit in } A \quad \forall y \in A\}$

Cor: Let $\mathfrak{m}$ be a maximal ideal & $1+x$ is a unit $\forall x \in \mathfrak{m}$, A is a local ring.

$$[\mathfrak{m} = J^n(A)]$$

Operation on ideals:

(*) $I+J := \{x+y : x \in I \text{ & } y \in J\}$

(*) $I \cap J$.

(*) IJ is not an ideal if we define this way $IJ := \{xy : x \in I \text{ & } y \in J\}$

(Ex: Find example)

(*) $IJ :=$ smallest ideal containing the set $\{xy : x \in I \text{ & } y \in J\}$.

If $I+J=A$, then ideals I & J are said to be ω -prime.

$$\begin{aligned}\varphi: A &\longrightarrow A/I \times A/J \\ a &\longmapsto (a+I, a+J)\end{aligned}$$

$$\ker \varphi = I \cap J$$

Suppose $I+J=A$,

$$\text{Let } (a+I, b+J) \in A/I \times A/J$$

$$\exists x \in I \text{ \& } y \in J \text{ s.t. } x+y=1.$$

$$bx = b - by$$

$$ay = a - ax$$

$$bx + ay = a + b - (by + ax)$$

$$\uparrow \qquad = a - ax + b(1-y)$$

Inspiration from chinese remainder thm

$\Rightarrow \varphi$ is surjective.

Ex: Converse is also true!!



φ is injective iff $I \cap J = \{0\}$

This can be extended to finitely many ideals.

⑩ If φ is surjective

$$\Rightarrow \varphi: A \longrightarrow A/I \times A/J$$

$$(a+I, b+J) \longleftarrow a$$

$$\Rightarrow a = a + i k \quad \text{for } i, k \in A.$$

$$a = b + j l$$

Claim: $I + J = A$ or $\exists r \in I$ & $s \in J$
s.t. $r + s = 1$

$$a + i k = b + j l$$

Now, take $a = 0, b = 1$

$$\Rightarrow \underset{\substack{\uparrow \\ I}}{i k} - \underset{\substack{\uparrow \\ J}}{j l} = 1 \quad \Rightarrow \boxed{I + J = A}$$